



Envision BESS Maintenance Technical Description

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Disclaimer

All technical data is subject to change in line with ongoing technical development. All the responsibilities and obligations are subject to the terms of the contract.

The illustrated figures in this documentation do not necessarily reflect the exact scope of supply. The actual scope of supply is subject to technical alterations at any time.

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1 Introduction

This document provides guidance and procedures on maintenance of the Battery Energy Storage System (BESS).

2 Safety



Safety is the prerequisite for all the site work.

To ensure the safety of operators and BESS equipment, all personnel involved in site work must carefully read and strictly abide by safety specifications. It is the worker's responsibility to meet all safety, local, and general regulations that apply for the specific site, task, or project.

For the specific safety requirements of different processes, refer to the related delivery documentation during contract performance phase.

3 Product Introduction

3.1 DC System

The DC system is a container-type centralized liquid-cooling system with high strength, strong load-bearing capacity, safety and reliability.

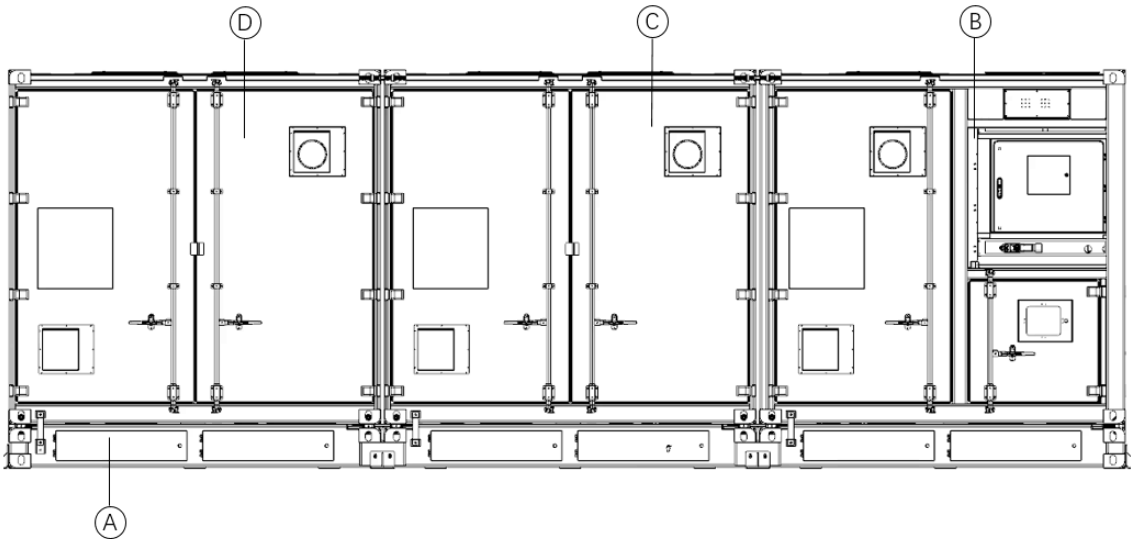


Figure 4-1 DC system

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No.	Name
A	SKID
B	Combined Segment
C	Battery Segment
D	Battery Segment



The figure applied here is for reference only. The actual product might vary based on different contracts.

3.2 AC System

The AC system includes equipment such as Power Conversion System (PCS), transformer, auxiliary transformer, and switch cabinet.

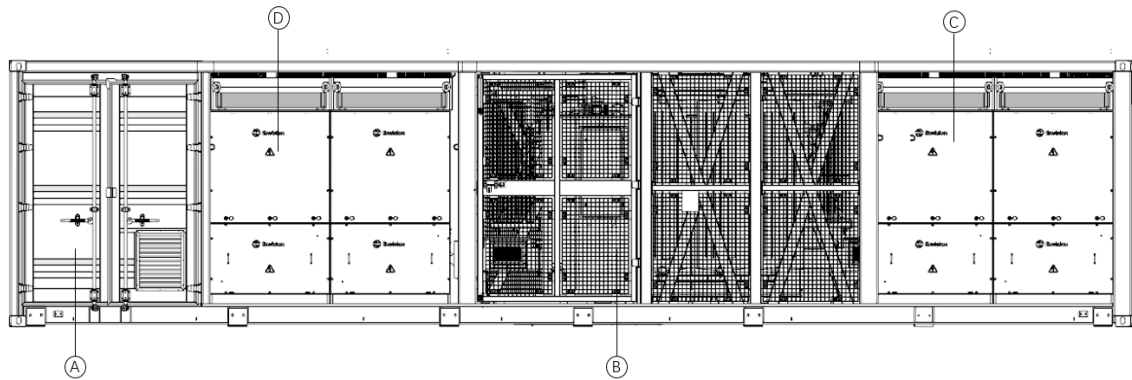


Figure 4-2 AC system

No.	Name
A	Switch cabinet
B	Transformer
C	PCS
D	PCS

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The figure applied here is for reference only. The actual product might vary based on different contracts.

4 Preparation

4.1 Preparation Before Maintenance

4.1.1 Equipment Power-Off

- 1) On the main control cabinet, turn the **Remote/Local** knob to the **Local** position.
- 2) Press the **Disconnect** button.
- 3) Disconnect the circuit breaker of the high-voltage (HV) box.
- 4) Disconnect all RACK-level contactors.



- Personnel must wear level-2 arc flash protection equipment when operating the power distribution cabinet.
 - For specific operations, see *Electrical Operating Safety Protection Manual for Energy Storage AC Integrated System*.
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4.1.2 Maintenance Cycle

Term		Description	Remarks
Maintenance cycle	A maintenance	After the on-site BESS equipment is powered on and connected to the grid, maintenance work is carried out every 6 months, and the execution period is ± 1 month.	After the on-site BESS equipment is powered on and connected to the grid, maintenance activities follow an alternating sequence of A maintenance and B maintenance.
	B Maintenance	After the on-site BESS equipment is powered on and connected to the grid, maintenance work is carried out every 12 months, and the execution period is ± 1 month.	The current maintenance cycle is carried out approximately 7 months after the theoretical time of the last maintenance. To ensure optimal performance, Envision recommends completing one full round of A maintenance and B maintenance within an actual interval of 5 to 7 months, finishing the entire maintenance cycle within 13 months.

5 DC System Maintenance

5.1 DC System Container Maintenance

5.1.1 DC System Container Exterior Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that the all door lock chains close properly, with no jamming or unusual noise. Replace them if any issues are found.
- 2 Visually inspect that the surface shows no signs of paint peeling or corrosion.
- 3 Visually inspect that the door seals are secure and reliable.
- 4 Visually inspect that the warning signs and labels are undamaged and clearly visible.

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- Promptly address any peeling protective paint exceeding 1 cm² on the surface.
- Replace door seals if there are issues with the door seal.
- Replace door locks promptly if they are not functioning properly.
- If there are dents on the door, see *Energy Storage Cabinet Anti-Corrosive Coating Repair Process*.

5.1.2 DC System Container Interior Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that there is no dust in the container. If any dust is found, use a vacuum cleaner to clean them.
- 2 Visually inspect that there is no rust or corrosion on any sheet metal components and connecting bolts. If any rust is found, apply zinc coating treatment or replace the affected parts as needed.
- 3 Visually inspect that all connecting bolts are not loose and the torque marking lines are not misaligned.



If a torque marking line is misaligned, re-tighten the bolt according to the torque list and mark the torque line accordingly.

5.1.3 Water-Cooling Unit Bottom Maintenance

Maintenance cycle: A maintenance, B maintenance

Maintenance requirements:

- 1 Visually inspect that there is no sand, dust, or other debris at the bottom of the water-cooling unit. If any dust is found, clean the container floor under the water-cooling unit using a high-pressure washer.

5.1.4 Cabinet Floor Drain Hole Inspection

Maintenance cycle: A maintenance, B maintenance

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Maintenance requirements:

- 1 Visually inspect that there is no standing water at the bottom of the cabinet. If any water is present, check the floor drain. If the floor drain is malfunctioning, remove any debris that might be obstructing the drain holes.



After cleaning, use a cloth to wipe away any remaining water stains and maintain cleanliness.

5.2 Grounding Weld Point Inspection and Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that all welding points are secure with no corrosion.
- 2 Visually inspect that all grounding bolts are not loose and the torque marking lines are not misaligned.

5.3 PACK Inspection and Maintenance

5.3.1 PACK Maintenance Requirements

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that the power cables and communication cables of the PACK are reliably connected with no arcing or blackening at the cable connectors.
- 2 Inspect that there is no leakage or corrosion on the PACK exterior.

5.3.2 PACK Battery Maintenance Requirements

Maintenance cycle: A maintenance, B maintenance

Maintenance requirements:

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- 1 Inspect that there is no imbalance in cell voltages within the PACK. If the battery cell charge is unbalanced, recharge the cells to achieve a balanced state.

5.4 HV Box Inspection and Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that there are no red indicator lights on the HV box.

Visually inspect the power cable connection terminal to check for any issues such as deformation or burning. Repair or replace the affected components if any issues are found.
- 2
- 3 Visually inspect that the grounding wires are not loose.

5.5 DC System Environmental Control Maintenance

5.5.1 Environmental Control Pipeline Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that there are no leaks in the water pipe joints of the PACK.
- 2 Visually inspect that there are no leaks in the clamps, pipe joints, quick connectors, and surfaces of the primary pipelines.
- 3 Visually inspect that there are no misaligned torque marking lines on the primary pipeline clamps.
- 4 Visually inspect that the nylon tube joints are intact and the exterior insulation of the nylon tube is complete and undamaged.



Use clamping torque tooling to tighten the pipeline clamps.

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5.5.2 Air Conditioner Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that the wiring connections are secure.
- 2 Inspect that the fan is operating properly with no unusual noise.
- 3 Inspect that the condenser is clean with no blockages. If it is blocked, clean the dust or lint-like debris on the surface of the condenser using a vacuum cleaner or air gun.

5.5.3 Water-Cooling Unit Inspection and Maintenance

Preparation before maintenance:

- 1 Unscrew the surrounding bolts and remove the surface cover plate of the water-cooling unit.

Maintenance cycle: A maintenance, B maintenance

Maintenance requirements:

- 1 Visually inspect that there is no rust on the external frames and sheet metal components. Apply touch-up painting or zinc coating if any rust is found.
- 2 Visually check the water-cooling unit and air inlet and outlet to ensure that there is no dust or foreign object. If any dust or foreign object is found, clean them.
- 3 Visually inspect that there is no dust or foreign object on the condenser. If any dust or foreign object is found, clean them.
- 4 Check the coolant level of the water tank. If it is below the H line, replenish the coolant to the H line.



Clean the condenser with compressed air or using a vacuum cleaner with a brush.

5.5.4 Water-Cooling Unit Cable Maintenance

Maintenance cycle: B maintenance

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Maintenance requirements:

- 1 Visually inspect that there is no aging or damage to the cables of the water-cooling unit.



Coordinate with spare parts for replacement if the external cable insulation is damaged.

5.5.5 Coolant Sampling Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Collect 50 mL of coolant and measure its freezing point using a freezing point tester, ensuring that the freezing point is below -35°C .
- 2 Collect 10 mL of coolant and measure its pH value, ensuring that the pH value is within the range of 7.5 to 8.5.
- 3 Collect 200 mL of coolant and submit it to a third-party agency for inspection.

5.6 Fire Protection System Inspection and Maintenance

5.6.1 Fire Alarm Function Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Use a smoke or heat gun to test the function of the fire alarm, ensuring that it can trigger an alarm.

5.6.2 Firefighting Device Start and Stop Button Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that the start button of the firefighting device works properly.
- 2 Inspect that the stop button of the firefighting device works properly.

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5.6.3 Aerosol Fire Extinguishing Agent Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Check that there is no leakage of aerosol fire extinguishing agent. If leakage occurs, replace the aerosol fire extinguishing device with a new one.

5.6.4 UPS Battery Health Inspection

Maintenance cycle: A maintenance, B maintenance

Maintenance requirements:

- 1 Check the UPS battery health.

5.6.5 UPS Battery Replacement

Maintenance cycle: every two years

Maintenance requirements:

- 1 Replace the UPS battery.

5.6.6 Combustible Gas Detector Maintenance

Maintenance cycle: A maintenance, B maintenance

Maintenance requirements:

- 1 Inspect that the combustible gas detector can trigger an alarm properly. If it cannot trigger an alarm, replace the combustible gas detector.

Maintenance cycle: every three years

- 1 Replace the combustible gas detector. Calibrate the combustible gas detector.



If the detector is replaced within the 3-year period, recalculate the maintenance cycle from the replacement date.

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5.6.7 Ventilation System Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that the fans are operating properly.
- 2 Inspect that there is no dust accumulated on the fans.



If any dust is found, clean it using a vacuum cleaner.

5.7 Power Distribution Cabinet Inspection and Maintenance

5.7.1 PDC Exterior Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that all door lock chains close properly with no jamming or unusual noise. Replace them if any issues are found.
- 2 Visually inspect that the surface shows no signs of paint peeling or rust. Promptly repair them if any issues are found.
- 3 Inspect that the door seals are reliable and the door locks are functioning properly.
- 4 Visually inspect that the warning signs and labels are undamaged and clearly visible.



- Promptly address any peeling protective paint exceeding 1 cm² on the surface.
 - Replace door seals if there are issues with the door seal.
 - Replace door locks promptly if they are not functioning properly.
 - If there are dents on the door, see *Energy Storage Cabinet Anti-Corrosive Coating Repair Process*.
-

5.7.2 Cable Gland Inspection

Maintenance cycle: B maintenance

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Maintenance requirements:

- 1 Visually inspect that the cable gland is locked securely and the torque marking line is not misaligned, ensuring that the cable is stable and free of gaps.
- 2 Visually inspect that the fireproof mud is intact and not missing.



If there is any loosening or gaps in the cable gland due to stress from cable tilt, adjust the cable to a vertical position and tighten the gland nut.

5.7.3 Electrical Connections Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that the power cables and communication cables inside the electrical cabinet are securely connected, with no signs of arcing or blackening at the cable connectors.
 - 2 Visually inspect that the connecting bolts on the copper busbar inside the PDC are not loose and the torque marking lines are not misaligned.
-



Before inspection, ensure that the BESS equipment is powered off, and the main switch and HV box are disconnected.

5.7.4 Cabinet Internal Components and Cables Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that there are no signs of arcing or corrosion on the components inside the cabinet.
 - 2 Visually inspect that there are no cracks or aging on the cable insulation and cable sheath.
 - 3 Visually inspect that the fixing bolts of the internal components are not loose.
-

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If the bolts are loose, tighten them according to the torque list and redraw the torque marking line.

5.7.5 UPS Battery Health Inspection

Maintenance cycle: A maintenance, B maintenance

Maintenance requirements:

- 1 Check the UPS battery health.

5.7.6 UPS Battery Replacement

Maintenance cycle: every two years

Maintenance requirements:

- 1 Replace the UPS battery.

5.7.7 Internal Cleaning

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that there is no dust or lint-like debris inside the cabinet. If dust or such debris is found, clean them.



Use a vacuum cleaner to remove the dust if any.

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6 AC System Maintenance

6.1 PCS Maintenance

6.1.1 Preparation Before Inspection

CAUTION

1. Only personnel who have been professionally trained and authorized by Envision are qualified to conduct the maintenance operations. The operations must comply with the *Electrical Operating Safety Protection Manual for Energy Storage AC Integrated System*.
2. Before starting maintenance, shut down the host computer. The two DC-side isolation switches (01-QS1 and 01-QS2) and ACB (01-Q1) within the PCS are automatically disconnected. Set maintenance switch 01-S1 to maintenance mode and remove the key.
3. Disconnect the isolation switch at the high-voltage side of the 35 kV transformer to ensure the low-voltage side of the transformer is powered off. Simultaneously, disconnect the two DC isolation switches of the battery main control cabinet to ensure that the PCS DC input is powered off.
4. Finally, if the auxiliary power is external power supply, it must be disconnected. If powered by UPS, disconnect the low-voltage-side UPS of the auxiliary transformer.
5. Before operation, adhere to procedures outlined in *Electrical Operating Safety Protection Manual for Energy Storage AC Integrated System* for power-off. Measure with a multimeter to ensure safety. Wear level-4 arc flash protection suit, insulated gloves, and safety shoes during measurement.
6. Before maintenance, use a multimeter to measure the grounding resistance of the PCS. Ensure that the resistance between the PCS main grounding busbar and the grounding point of the integrated system base is less than or equal to 1 Ω .
7. Ensure that the power-off period exceeds 30 minutes before equipment maintenance. Use a multimeter to measure voltages on capacitors, copper busbars, and so on, ensuring that they are below the safe level before continuing maintenance.
8. Clean dust with a vacuum cleaner. Avoid using blowing tools to prevent short circuits.
9. Be careful when removing the door panel during PCS maintenance to prevent injury.
10. During maintenance, prioritize attention to warning labels within the PCS, emphasizing awareness of component hot surfaces. Certain cabinet elements, such as power modules, capacitors, radiators, copper busbars, inductors, cables, and heat exchangers, retain elevated temperatures for a duration after the PCS is powered off.
11. After disconnecting the DC isolation switch and ACB, residual voltages of 1500 V DC or 690 V AC persist in specific sections. For inspection and maintenance involving these areas, including circuits related to circuit breaker incoming sides, measure the voltage with a multimeter beforehand to ensure it is zero. During measurement, adhere to safety protocols, wearing level-4 arc flash protection suit, insulated gloves, and safety shoes. Follow the provided table below to disconnect the corresponding switches for

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maintenance on other circuits inside the PCS.

12. If communication is poor after logging in to the host computer and plugging or unplugging optical fibers is necessary, securely grip the fiber connector rather than the fiber itself. Avoid direct contact with the fiber end with bare hands to prevent potential damage to the fiber.
13. The minimum bending radius for optical fibers must exceed 15 times the diameter. Overly bending the fibers might result in damage.
14. To maintain other circuits in the PCS, disconnect the corresponding switches according to the table below.

+POW-01-F4	PCS pre-charge circuit
+POW-01-F4/01-F5	Voltage detection circuit for energy storage circuit breaker (input and output)
+POW-16-Q0	PCS 400 V power supply
+POW-16-F0	PCS 230 V UPS power supply
07-F1 in external water-cooling cabinet	Water-cooling cabinet 400 V power supply
07-F2 in external water-cooling cabinet	Water-cooling cabinet 230 V UPS power supply
+POW-16-Q1	Power cabinet reactor/kit fan protection
+POW-16-Q2	Grid cabinet fan protection
	Water-cooling cabinet fan protection

6.1.2 PCS Cabinet Inspection and Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that the cabinet door is reliably sealed.
- 2 Visually inspect that the cabinet surface shows no signs of rust or damage.

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- Promptly address any peeling protective paint exceeding 1 cm² on the surface.
- If there are dents on the door, see *Energy Storage Cabinet Anti-Corrosive Coating Repair Process*.
- Replace door seals if there are issues with the door seal.

6.1.3 Fireproof Mud Inspection and Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect and confirm that the fireproof mud around the copper busbar is intact. If any fireproof mud is missing, reapply it properly.
- 2 Inspect that the fireproof mud seal at the gap of the power cable gland is sealed properly.



If any looseness or gaps are found in the cable gland due to cable strain from tilting, adjust the cables to be vertical, tighten the gland fittings, and properly seal with fireproof mud.

6.1.4 Power Module and Busbar Connection Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that all visible main circuit bolts show no signs of corrosion.
- 2 Visually inspect that the bolts are not loose and the torque marking lines are not misaligned.
- 3 Visually inspect that the cables and bolt joints show no signs of overheating, blackening, or sparking.



If the bolts are loose, tighten them according to the torque list and redraw the torque marking line.

6.1.5 Main Circuit Cable Connections Visual Inspection

Maintenance cycle: B maintenance

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Maintenance requirements:

- 1 Visually inspect that the torque marking lines of the main circuit bolts on the grid side, motor side, and so on, are not misaligned. If any misalignment is found, use a torque wrench to verify the torque of all bolts.
- 2 Inspect that the bolts show no rust on the exterior.



- If the bolts are loose, tighten them according to the specified torque value.
- If severe rust is found, replace the bolt with a new one.
- Reapply a torque marking line and anti-rust paint to the bolts that have been tightened to the specified torque.
- Ensure no signs of heating or blackening on the cable joints.

6.1.6 Cabinet Internal Electrical Connections Visual Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect and ensure no damage to the outer sheath of cables at the power cable outlet, and no discoloration or rust at the cable connecting points.
- 2 Inspect that the clamps and connectors used to secure cables inside the cabinet are in good condition.
- 3 Inspect that the torque marking lines on the fixing bolts of grounding points on the main grounding busbar are not loose.



- If any damage is found, re-insulate or replace them.
- If the cables are loose, re-tighten them.

6.1.7 Main Circuit Breaker Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Ensure that the entire system safety chain is engaged before switching the PCS to PC mode and closing the circuit breaker on the **Converter Manager** interface.
- 2 Press the emergency stop switch on the PCS cabinet door, and verify that the circuit

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breaker is disconnected normally.

6.1.8 Auxiliary Device Bolt Wiring Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Check the tightness of wiring, and conduct a random sampling of 20 wiring points. In case of non-compliance in the sampled points, perform a thorough inspection and tighten all connections.
- 2 Visually inspect that the torque marking lines on the bolts are not misaligned. If any bolts are misaligned or loose, manually tighten them with an insulated screwdriver after powering them off.



Perform a comprehensive inspection and tighten all if any are loose.

6.1.9 Filter Capacitor Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect the filter capacitors to ensure that there are no bulges, damage, or raised top covers. Additionally, inspect that there are no signs of burning or discoloration on the connecting terminals and cables.

6.1.10 Internal Water-Cooling System Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect all pipes, pipe joints, quick connectors, and radiator core surfaces to ensure no signs of leakage.
- 2 Verify that there are no wrinkles or aging on the surface of the rubber hoses and no cracks at the hose clamps.
- 3 Ensure that there is no dust or blockages to the internal radiator. If they are present, clean them using a brush.

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4 Activate the fans using the software and ensure that there is no unusual noise during operation.

5 Inspect that the positions marked on all clamping joints are aligned.



- The cabinet has 9 pipe branches, with 6 for the kit, 2 for the reactor, and 1 for the control cabinet. All hose joints must be torqued once a year.
- Replace the hoses if any wrinkles or cracks indicating aging are found during the visual inspection.
- For liquid filling procedures, see *3.45MW Energy Storage AC Side Liquid-Cooling System Liquid Filling and Air Venting Operation Guide*.

6.1.11 Dehumidifier Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that the drainage pipe joint is secure, the dehumidifier drainage pipe is straight with no entanglement or knots to facilitate drainage.
- 2 Inspect that the drainage pipe shows no signs of leakage.

6.2 Transformer Maintenance

6.2.1 Transformer Appearance Inspection Requirements

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that the conductive parts show no signs of rust, corrosion, electrical tracking, or carbonization on insulating surfaces.
- 2 Check the reliability of fasteners, grounding wire connections, and conductive parts to ensure no rust, discoloration, or corrosion traces.
- 3 If excessive dust is found, it must be removed.
- 4 Ensure that the power cable connections on the high-voltage and low-voltage sides are secure.

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- Do not use a wet cloth to wipe any components or parts.
- Use a vacuum cleaner to clean significant dust if any.

6.2.2 Transformer Oil Level Inspection Checklist

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Ensure that the transformer oil level is within the normal range.

6.2.3 Desiccant Oil Cup Silica Gel Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that there is oil in the transformer oil cup.
- 2 Ensure that the desiccant (anhydrous silica gel) in the dehumidifier is not saturated with water.

6.2.4 Transformer Oil Inspection Requirements

Maintenance cycle: once every two years

Maintenance requirements:

- 1 Ensure that the transformer oil test results are normal.

6.3 Low-Voltage Cabinet Maintenance

6.3.1 Low-Voltage Cabinet Interior Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that the components show no signs of rust or corrosion.

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- 2 Inspect the cables to ensure no signs of loosening.
- 3 Inspect the battery casing and cover to ensure no leakage, deformation, cracks, or damage.
- 4 Inspect the bottom cable glands to ensure that they are tightened securely with no loosening.
- 5 Inspect the fan fixing bolts inside the low-voltage cabinet to ensure that they are tightened securely with no loosening.
- 6 Inspect that the fixing bolts on the door heater are tightened securely.



During B maintenance, verify that the heater and fans are working properly.

6.3.2 UPS Battery Health Inspection

Maintenance cycle: A maintenance, B maintenance

Maintenance requirements:

- 1 Check the UPS battery health.

6.3.3 UPS Battery Replacement

Maintenance cycle: every two years

Maintenance requirements:

- 1 Replace the UPS battery.

6.4 High-Voltage Room Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visual inspect that there are no signs of electrical discharge or physical damage, and ensure that there are no indications of overheating on the contact surfaces.
- 2 Visually inspect that the casing shows no signs of deformation, peeling, rust, and water

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leakage. Confirm that bolts and connectors are securely fastened to the base frame.

- 3 Inspect the high-voltage input terminal connection for sufficient overlap with the copper busbar, and ensure that the bolts are securely fastened with no loosening.
- 4 Inspect the clamps for high-voltage input cables to ensure that they are not loose.
- 5 Inspect the door lock to ensure that it is locked effectively and has no rust.
- 6 Inspect the grounding wires to ensure that they are securely connected with no loosening, and that the connection points show no signs of rust, discoloration, or corrosion.
- 7 Inspect that there are no cracks or dirt accumulation on the surface of the lightning arrester.

6.5 Environmental Control System Maintenance

CAUTION

1. Before operation, power off the system and perform LOTO to prevent accidental operations that might cause personal injury. Power-on operations must be performed by qualified personnel.
2. Be cautious of high temperature (70°C–80°C) around water pipes and drive motors to avoid burns.
3. Do not add pure water or other brands of coolant to the water-cooling system.
4. The coolant must be stored in dedicated bottles or barrels, and appropriate labels must be affixed to their surfaces.
5. When performing pressure relief or air venting operations, take precautions against coolant splashes and wear masks to prevent coolant from entering your mouth or nose.
6. Do not randomly adjust the valve positions that have already been set in the system.
7. Do not disassemble or assemble pipelines or valve bodies under pressure.

6.5.1 Radiator Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that there is no damage, deformation, or corrosion on the radiator.
- 2 Visually inspect that the radiator is properly sealed with no leaks.

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- 3 Visually inspect that there are no foreign objects in the air duct.
- 4 Verify that the radiator fan operates with no unusual noise.
- 5 Visually inspect that there are no leaks or rust on the exhaust valve of the pipeline.
- 6 Use a high-pressure washer to remove the accumulated dust from the radiator.



- Clean any dust and debris from the radiator fins.
- If any rust is present, remove it and apply anti-rust oil.

6.5.2 Coolant Sampling Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Use a freezing point tester to check the freezing point of the coolant, which must be below -30°C .
- 2 Measure the pH value of the coolant, which must be within the range of 7.5 to 8.5.
- 3 Collect 200 mL of water coolant and submit it to a third-party agency for testing.



- For the AC system, conduct coolant sampling with a random inspection ratio of 10%.
- For the reference process see *3.45MW Energy Storage AC Side Liquid-Cooling System Liquid Filling and Air Venting Operation Guide*.

6.5.3 Coolant Pressure Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that the water coolant pressure is above 1.5 bar. If the pressure is insufficient, replenish to 1.5 bar.



For the reference process, see *3.45MW Energy Storage AC Side Liquid-Cooling System Liquid Filling and Air Venting Operation Guide*.

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6.5.4 Water-Cooling Pump Station Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that there is no damage, deformation, or corrosion on the pump station appearance.
- 2 Visually inspect that the pump station is properly sealed with no leaks.
- 3 Ensure that there is no abnormal vibration or unusual noise during operation.

6.5.5 Expansion Tank Maintenance

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that the manual exhaust valve at the top shows no leakage.



- Inspect the manual exhaust valve on the top of the expansion tank. If any leakage is found, tighten it with torque.
 - If the internal bladder of the expansion tank is ruptured or the gas valve of the expansion tank is malfunctioning, replace them. For detailed procedures, see *PRC-0013206 3.45MW Energy Storage AC Side Liquid Cooling System Operation and Maintenance Process Specification*.
-

6.5.6 Heater Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Visually inspect that the heater exterior shows no damage, deformation, corrosion, and so on. Ensure good sealing with no leaks, and apply thread-sealing tape.
- 2 Connect the heater to the host computer and start it up to observe if the water temperature rises as expected.



Thread specifications:

- Terminal block M4
 - Junction box cover M5
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- 2-inch pipe thread
-

6.5.7 Instrument Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that temperature sensors, pressure sensors, pressure gauges, and automatic/manual exhaust valve have tight sealing with no leaks, and the thread is wrapped with sealing tape.
- 2 Inspect that there is no damage, deformation, or corrosion on the appearance.

6.5.8 Pipelines and Joints Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that the pipelines and joints have tight sealing with no leaks.
- 2 Inspect that there is no damage, deformation, or corrosion on the appearance.



Use a soft cloth to wipe or wear gloves and touch to check for water marks.

6.5.9 Water-Cooling System Torque Inspection

Maintenance cycle: B maintenance

Maintenance requirements:

- 1 Inspect that the clamp fastening bolts are not loose as required. If any misalignment is found, tighten them all with torque.

6.6 Structural Assembly Maintenance

6.6.1 Structural Assembly Exterior Maintenance

Maintenance cycle: B maintenance

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Maintenance requirements:

- 1 Inspect that all door lock chains close normally with no jamming or unusual noise. If any issues are found, replace them.
- 2 Visually inspect that the surface shows no signs of paint peeling or corrosion. If any issues are found, repair them promptly.
- 3 Inspect that the door is sealed securely.
- 4 Visually inspect that the warning signs and labels are clearly visible, undamaged and clean.
- 5 Visually inspect that there is no dust inside. If any dust is found, clean it using a vacuum cleaner.
- 6 Visually inspect that the bolts and sheet metal parts inside the cabinet are free of rust. If any rust is found, perform zinc spraying.
- 7 Inspect that the grounding bolts are not loose and the torque marking line is not misaligned.



- Promptly address any peeling protective paint exceeding 1 cm² on the surface.
- If there are dents on the door, see *Energy Storage Cabinet Anti-Corrosive Coating Repair Process*.
- Replace door seals if there are issues with the door seal.
- Replace the door locks promptly if they are not functioning properly.

6.6.2 Structural Assembly Ventilation Maintenance

Maintenance cycle: A maintenance, B maintenance

Maintenance requirements:

- 1 Inspect that the air inlet is not blocked. If it is blocked, clean it with a vacuum cleaner.
- 2 Inspect that the air outlet is not blocked. If it is blocked, clean it with a vacuum cleaner.
- 3 Inspect that the fan is operating properly with no unusual noise.
- 4 Check that the fan fixing bolts are tightened securely.

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- If any dust is found, clean it using a vacuum cleaner.
 - Ensure that the filter cotton in the vent is replaced once every six months.
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